Summary Python (Exam-Content)

Introduction to python (Universiteit van Amsterdam)

Formula's used in past exams Python

Basic Logic
Integer -> 123
Float -> 123.3
String -> '123'
Tuple -> ('123', 123)

Can compare integer & float
Can't compare string with a number

Selecting from tuples:
x[0][1] --> means take the 0^{th} element in X, then take the 1^{st} element in that tuple
i.e. [0] would indicate take (1,2), [1] would indicate = 2

Basic Scope
If variable is defined locally (indented) cannot be referenced outside of function
Global variables are in the outermost scope (not inside any function)

Basic Lists
[[output expression] for (iterator variable) in (iterable)]

OOP
```
class MyClass
        def _ _ init _ _(self, var1, var2 = " "):
                self. var1 = var1
                self. var2 = var2
```

var2 = " " only if it needs to be specified at object construction

Basic Loop
Range (0, X) doesn't include X, i.e. Range(0, 20) is from 0 – 19

Strings
F-string
f"literal string {expression}"

i.e.
```
for list in lists:
        print(f"{list['key1']} has blahblah {list['key2']:.(rounding of number)f}."
```

Replace
Need: number = number.replace
Can't do s1.replace( )

Pandas
Summing

df[“C”] = df[“A”] + df[“B”]

df.loc [row, column]
-> can look up with integer (i.e., 2) or with name (i.e., “B”)

Ascending = True is default

Vectorize string operation
s.map(lambda x: x[(number):(number):(number of steps)]

Dictionary Comprehension
{<output expr.> for <it.var> in <iterable>}
Iterable = list
Comparing dictionaries in python:
D1 == D2: True if they contain the same values (not just a specific order) i.e. {1: 5, 4: 6} = {4:6, 1: 5}

Trial
Dictionary:
d1 = { }
for key, value in zip(L1, L2)
    d1[key] = value
Maps every value of List 1 to List 2

d2 = { }
for key, value in enumerate(L2, 1)
    d2[key] = value
Maps indexes (0,1,2, etc..) to L2

For key, value in d1.items()

- Can’t just have value in d1:

Functions
String.count(substring) to count

Examples to Remember

All of the following will create a list multiplying every element of x by 2:

1. list(map(lambda i: i * 2, x))
2. [(lambda i: i * 2)(item) for item in x]
3. y = lambda i: i * 2 [y(j) for j in x]

Creating a new list where elements are equal to their indices

```
y = []
for i, j in enumerate(x)
if i == j:
      y.append(j)

could also be j. append(i)
```

```
y = [i for i in x if i == x.index(i)]
```

Grades Weekly Programming
1: 10
2: 8.88
3: 8.88
4: 10
5: 8.88

= 9.328

Weekly Programming Average: 93.28%
Midterm: 5
Endterm: 6

Grade weekly programming with week 3 = 0:
7.773